

MIDWEST GENETICS LABORATORY

Clinical Molecular Genetics

840 S. Wood Street, Chicago, IL 60612

Phone: (312) 555-6100 | Fax: (312) 555-6109 | CLIA #: 14D1022548

CLINICAL GENETIC TESTING REPORT

Patient: Marcus D. Whitfield **MRN:** PED-2025-04471
DOB: 03/03/2016 **Age:** 9 years, 2 months **Sex:** Male
Ordering Provider: Anita Sharma, MD — Cook County Pediatrics
Specimen Collected: 05/15/2025 **Report Date:** 06/01/2025
Specimen Type: Peripheral blood (EDTA)
Test Ordered: NF1 Gene Sequencing and Deletion/Duplication Analysis

INDICATION FOR TESTING

Clinical suspicion for Neurofibromatosis Type 1 based on multiple cafe au lait macules (reported as 7 or more, each greater than 0.5 cm) and axillary freckling noted on physical examination.

RESULTS

Variant Identified: NF1 c.2033A>G (p.Asn678Ser)
Classification: Variant of Uncertain Significance (VUS)
Zygosity: Heterozygous

INTERPRETATION

A heterozygous missense variant was identified in the NF1 gene (c.2033A>G, p.Asn678Ser). This variant results in an asparagine-to-serine substitution at position 678 in the neurofibromin protein. This variant has not been previously reported in the literature or in large population databases (gnomAD, ClinVar) at the time of this report. In silico predictive tools show conflicting results. Based on current evidence, this variant is classified as a Variant of Uncertain Significance (VUS) per ACMG/AMP guidelines (Richards et al., 2015).

A VUS should not be used alone to make clinical decisions. However, the patient's clinical presentation (multiple cafe au lait macules and axillary freckling) meets NIH diagnostic criteria for NF1 independently of genetic testing. Clinical management should be guided by the clinical diagnosis rather than this genetic result alone.

RECOMMENDATIONS

- Clinical NF1 diagnosis can be made independently of this genetic result if ≥ 2 NIH criteria are met.
- Periodic reclassification of this variant may occur as new evidence becomes available. The ordering provider and patient/family will be notified if the classification changes.
- Genetic counseling recommended to discuss implications for the patient and family.
- Ophthalmologic evaluation recommended to screen for Lisch nodules and optic pathway glioma.

Laboratory Director: Patricia Holden, MD, PhD, FACMG

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